Pair of Linear Equations in Two Variables

The big idea

An equation like $2x + 3y = 12$ has infinitely many solutions, and its graph is a straight line. A **pair** of such equations asks: which (x, y) satisfies **both** at once? Geometrically, that is where the two lines meet.

THREE POSSIBILITIES

Two lines in a plane either **intersect** at one point (one solution), are **parallel** (no solution) or **coincide** (infinitely many solutions).

Checking consistency without drawing

For $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$, compare the ratios:

Comparison	Lines	Solutions	Pair is
$a_1/a_2 \neq b_1/b_2$	Intersecting	Exactly one	Consistent
$a_1/a_2 = b_1/b_2 = c_1/c_2$	Coincident	Infinitely many	Consistent (dependent)
$a_1/a_2 = b_1/b_2 \neq c_1/c_2$	Parallel	None	Inconsistent

IS THE PAIR X + 2Y − 4 = 0 AND 2X + 4Y − 12 = 0 CONSISTENT?

$a_1/a_2 = 1/2$, $b_1/b_2 = 2/4 = 1/2$, $c_1/c_2 = -4/-12 = 1/3$.
The first two ratios are equal but differ from the third, so the lines are **parallel**: no solution, **inconsistent**.

Graphical method

1. For each equation, find two or three points (make a small table of x and y values).
2. Plot both lines on the same axes.
3. Read the coordinates of the intersection point.

This works well for neat whole-number answers but is slow and imprecise otherwise, which is why we also learn algebraic methods.

Substitution method

1. From one equation, express one variable in terms of the other, e.g. $y = 7 - x$.
2. Substitute into the other equation to get one equation in one variable.
3. Solve, then substitute back.

SOLVE X + Y = 14 AND X − Y = 4

From the first, $y = 14 - x$. Substitute: $x - (14 - x) = 4 \Rightarrow 2x = 18 \Rightarrow$ **x = 9**, so **y = 5**.

Elimination method

Multiply the equations so that one variable has equal (or opposite) coefficients, then add or subtract to eliminate it.

SOLVE 3X + 4Y = 10 AND 2X − 2Y = 2

Multiply the second by 2: $4x - 4y = 4$. Add to the first: $7x = 14 \Rightarrow$ **x = 2**.
Then $2(2) - 2y = 2 \Rightarrow$ **y = 1**.

WORD PROBLEMS

Always define your variables first ("Let the cost of one pen be ₹x…"), turn each sentence into an equation, solve, and **answer in words with units**.

SIGN SLIPS WHILE SUBTRACTING

When subtracting equations, change the sign of **every** term of the second equation, including the constant.

Quick recap

- Each linear equation is a line; the solution of the pair is the meeting point.
- Use the ratios a_1/a_2, b_1/b_2, c_1/c_2 to decide: one, none or infinitely many solutions.
- Algebraic methods: substitution and elimination.

Practice questions

1. The pair $x + y = 5$ and $2x + 2y = 10$ has
   (a) no solution (b) a unique solution (c) infinitely many solutions (d) exactly two solutions
2. For what value of k will $3x + y = 1$ and $(2k - 1)x + (k - 1)y = 2k + 1$ have no solution?
   (a) 1 (b) 2 (c) 3 (d) –2
3. Solve $x + y = 14$ and $x - y = 4$.
   (a) $x = 10, y = 4$ (b) $x = 9, y = 5$ (c) $x = 8, y = 6$ (d) $x = 7, y = 7$
4. Two lines are parallel. The pair of equations is
   (a) consistent (b) inconsistent (c) dependent (d) always solvable
5. If $2x + 3y = 7$ and $3x + 2y = 8$, then $x + y$ equals
   (a) 2 (b) 3 (c) 4 (d) 5

Answers: 1. c 2. b 3. b 4. b 5. b